THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Chloride ion is: 1

(a) Cl^{2-} **(b)** Cl^{+} **(c)** Cl^{-} **(d)** Cl^{2+}

2. Covalent compounds usually have: 1

(a) high conductivity **(b)** high melting points
(c) low melting points **(d)** hard crystals

3. Dalton proposed: 1

(a) nucleus **(b)** periodic table **(c)** electron **(d)** atomic theory

4. Oxide ion is: 1

(a) O^{2+} **(b)** O^{2-} **(c)** O^{+} **(d)** O^{-}

5. Atomic mass unit is written as: 1

(a) N **(b)** g only **(c)** u **(d)** kg only

6. A molecule of oxygen has: 1

(a) 4 atoms **(b)** 2 atoms **(c)** 3 atoms **(d)** 1 atom

7. The formula of sodium chloride is: 1

(a) $NaCl$ **(b)** Na_2Cl_2 **(c)** $NaCl_2$ **(d)** Na_2Cl

8. The atomic mass of carbon is: 1

(a) 14 u **(b)** 12 u **(c)** 6 u **(d)** 16 u

9. A molecule of ozone has: 1

(a) 1 atom **(b)** 4 atoms **(c)** 2 atoms **(d)** 3 atoms

10. The formula of magnesium oxide is: 1

(a) MgO_2 **(b)** MgO **(c)** Mg_2O **(d)** Mg_2O_3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is a cation? Give an example. 2

12. Write the formula of water and carbon dioxide. 2

13. Define a molecule. 2

14. What is an anion? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the molecular mass of NH_3 and CH_4. 3

16. Name three compounds and write their formulae. 3

17. Write the symbols of the ions: chloride, oxide, nitrate. 3

OR

Calculate the molecular mass of CO_2 and SO_2 (S = 32). 3

18. Give three examples of molecules of elements. 3

OR

State three differences between atoms and molecules. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe Dalton's atomic theory. 5

OR

Explain ions with examples of cations and anions. 5

20. Explain the formation of NaCl. 5

OR

Write about atoms and molecules and give the atomicity of five substances. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND